

Reflecting on Memory Patterns in the Post-Human Era: AI, Digital Space, and the Reconfiguration of Embodied Memory

Abstract

This essay explores how artificial intelligence (AI) redefines human memory mechanisms in the contemporary post-human era by reconfiguring spatial and embodied experiences. Situated at the intersection of architecture, philosophy, and media theory, the research draws on Plato's concept of anamnesis and Bernard Stiegler's theory of three types of memory to analyse the transition from memory rooted in sensory and bodily experience to one increasingly mediated by AI-generated digital simulation.

This shift highlights a fundamental tension between two modes of memory: the immediacy of lived experience and spatial perception emphasised in phenomenology, and the algorithmic reconstruction and storage of such experiences by AI systems. The essay argues that this digital transformation raises philosophical questions concerning authenticity, memory sovereignty, and epistemic agency.

In this context, physical and virtual spaces are increasingly co-producing our modes of recollection. Memory is no longer merely a subjective impression within the mind or a residual trace of bodily experience—it has become a computational, retrievable, and reproducible spatial phenomenon. As AI becomes more deeply involved in constructing space and generating memory, the intimacy and embodiment of remembering grow increasingly fragile. This phenomenon compels a renewed reflection on the modes of memory's existence and expression in the digital age.

Keywords: Artificial Intelligence, Memory Reconstruction, Embodiment, Phenomenology, Post-Human Era, Digital Simulation, Virtual Space

Introduction

In ancient Greek philosophy, Plato's concept of Anamnesis refers not merely to recollecting past experiences but to reawakening truths already inscribed in the soul (Plato, 1972). For Plato, memory is not a form of "storage," but one of "representation" or "awakening"; learning is not the accumulation of information, but the act of "remembering" knowledge that already exists within the soul (Sedley, 2010).

From ancient times, memory has remained central to human cognitive experience. In early human societies, memory was transmitted through oral traditions, rituals, and symbolic spaces. With the advent of writing systems, memory became "externalised" into text. This development was further accelerated by the mechanical reproduction of images and sound via printing, photography, and phonographs. These technologies expanded the boundaries of memory and subtly reshaped its definition: memory was no longer solely an internal, subjective perception, but increasingly became information that could be preserved, transferred, shared, and even manipulated.

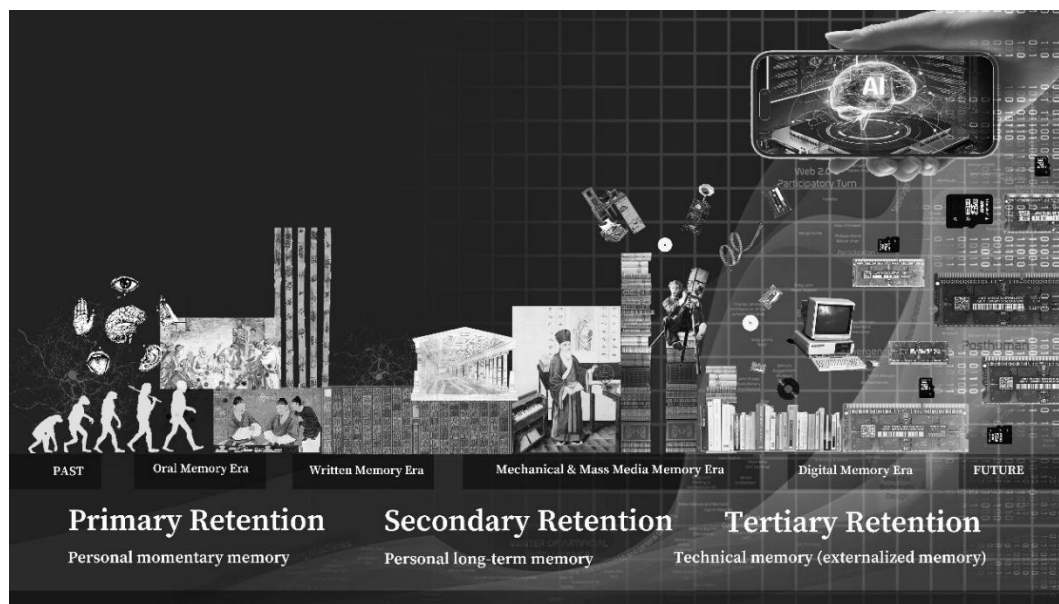


Figure 1 The Evolution of Memory and Stiegler's Three-Tiered Theory

Source: Author.

Bernard Stiegler's theory of three forms of memory provides a philosophical response to

this evolution: he categorises memory into primary memory (epiphylogenesis), which is biological; secondary memory (hypomnesis), which is social; and tertiary retention, or external memory, which is technological (Stiegler, 1998). With AI's rapid advancement, external memory's influence has become increasingly pronounced; humanity is entering a post-human era in which memory becomes algorithmic, editable, and even generatively "reconstructible."

AI's capabilities in image recognition, narrative generation, and virtual spatial construction are also redefining the relationship between memory and space. In traditional contexts, memory is deeply rooted in embodied experience and physical environments, such as cultural landmarks, daily life scenes, and is closely related to sensory perception. However, memory is increasingly transformed into digital material that can be manipulated, reconfigured, and even virtually created.

This leads to the core question of this essay: How does artificial intelligence reconstruct space to intervene in and reshape the relationship between human beings and memory?

Memory and Space: From Embodied Experience to Spatial Mnemonics

As emphasised in Plato's theory of anamnesis, memory was originally an internal cognitive process closely tied to individual experience. Throughout human history, the generation and reproduction of memory have always been inseparable from space, not only because space provides sensory stimuli and psychological orientation, but also because the body itself participates in the process, forming the foundation of what is known as embodied memory.

Before the invention of writing, oral traditions, ritualistic narratives, and cave paintings served to preserve and transmit collective memory. Memory relied heavily on this stage's tight linkage between sensation, movement, and spatial context. Techniques such as the

ancient Greek Method of Loci and Matteo Ricci's "memory palace" method illustrate how people constructed familiar spatial frameworks in their minds, embedding abstract information within specific spatial nodes to achieve more stable recall (Spence, 1985). Here, space functioned as a container and a structural framework through which memory occurred.

With the advancement of technology, memory gradually underwent a shift toward exteriorisation. The invention of the printing press enabled the mass reproduction and circulation of knowledge, detaching it from individual memory. The emergence of photography and phonographs further transformed memory into visible, audible, and storable forms, freeing it from the reliance on the body and space. These technologies marked a turning point, where memory began to be projected onto physical media, abstracted away from embodied experience.

In the digital age, hard drives, cloud storage, databases, and algorithms have constructed vast information storage and retrieval systems. The spatiality of memory has been further abstracted into data structures and encoding protocols. What Stiegler refers to as tertiary memory has now reached an unprecedented scale, accelerating the circulation and accessibility of information while further stripping memory from human experiential dimensions.

With the intervention of artificial intelligence, memory is no longer merely stored but actively processed: AI can recognise images, reconstruct narratives, simulate memory fragments, and even participate in the reconfiguration of the past. Memory no longer belongs solely to the individual but emerges as a hybrid mechanism of human-machine co-production, interaction, and shared generation.

This leads to a critical philosophical inquiry: Can memory retain its emotional depth and existential authenticity when detached from sensory experience and the spatial background embedded in the body? From the perspectives of embodiment and

phenomenology, the body and senses are not only channels of memory but also sources of emotional resonance. Once screen images and data models replace the embodied dimension of memory, what is evoked—true recollection or a simulation of the past? In this process, traditional notions of embodied cognition face unprecedented challenges. As Gangopadhyay and Pichler (2024) suggest, “In everyday digital technological environments, our agency must be distinguished between embodied and disembodied agency, and how the two co-evolve through digital interaction remains an open question.”

Memory in Digital Reconstruction: How Space is Reproduced and Perceived

As previously discussed, whether embodied memory within oral traditions or externalised memory enabled by technological media, space has always played a crucial role in generating and storing memory. This reliance on “place” goes beyond geographic location, reflecting a deep psychological and emotional attachment. When we say we “remember a place,” we often refer to the feelings, relationships, and events experienced in that spatial context.

At the urban scale, cultural landmarks, commemorative architecture, and neighbourhood structures are anchors of personal memory and carriers of collective memory (Aldo, 1982). For example, war memorials embody emotional experiences of mourning and shared identity. Similarly, preserving old streets and buildings contributes to generational perceptions of home or childhood. These physical environments become “memorised” through sustained bodily engagement, communal activity, and emotional investment, forming a crucial foundation for cultural continuity.

With technological advancement, the preservation and transmission of spatial memory have gradually shifted toward digital realms. Virtual museums, online memorials, 3D mapping, and AI-based reconstruction projects enable spaces that have disappeared—or are disappearing—to be “re-presented” in new ways. These tools overcome physical

limitations, allowing wider access to distant or lost memories. Yet, they also raise critical questions: Can digitally reconstructed spaces evoke the same emotional resonance and sense of reality as their physical counterparts?

Take the example of Hong Kong Kowloon Walled City, since its demolition in 1994, it has lived on in public memory as a potent symbol of collective urban history. As Fraser and Li (2017) argue, the “real” memory of Kowloon Walled City as a physical site has been replaced by a culturally constructed memory, which now exists at the intersection of reconstruction and emotion. In recent years, the City’s image has been repeatedly reproduced in documentaries, virtual museums, video games, and immersive experience platforms. The team from the Hong Kong University of Science and Technology used virtual reality (VR) technology to reconstruct the interior spaces of the walled city, allowing users to “walk” through it (Hong Kong University of Science and Technology, 2024). This digital reconstruction detaches from the physical site yet strives to evoke the collective perception and memory of the vanished space.

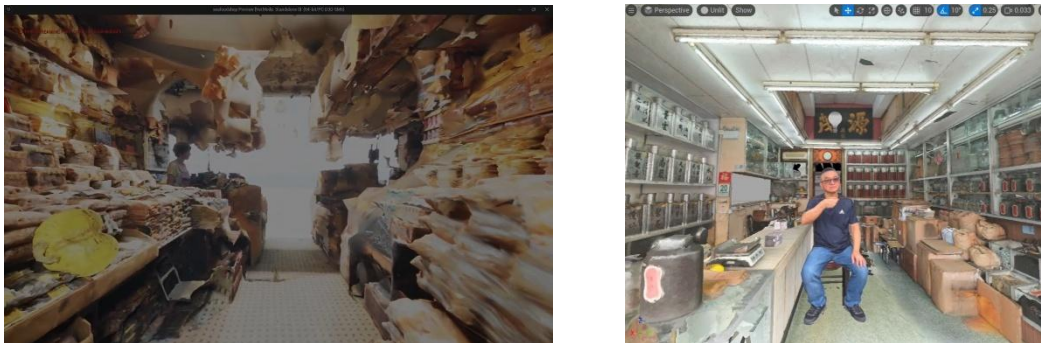


Figure 2&3: Kowloon City’s old shops VR project - HKUST

Source: Digital Storytelling with VR: Visual Conservation of Kowloon City’s Heritage - HKUST Digital Humanities Initiative

Other notable examples of digital spatial reconstruction include the work of Condell *et al.* (2021), who created a virtual museum and 3D reconstruction of a 16th-century church through community collaboration, and Li (2022), who built a virtual scene of Basha Village using VR to preserve ethnic cultural heritage.

In these simulations, memory is translated into visual and interactive experiences. But this also raises concerns: Do these digital reconstructions retain the sensory richness and emotional depth of original lived experience? Returning to the core of embodiment and phenomenology: memory is not merely “seen,” but “felt” (Merleau-Ponty, 2012). As digital spaces trend toward perfection and control, we may be losing the most intimate, uncertain, and irreplaceable fragments of memory.

Memory with Artificial Intelligence: From Assistance to Algorithmic Co-Creation

If digital space offers new containers and memory settings, artificial intelligence intervention propels the shift from static storage to dynamic construction and re-enactment. Artificial intelligence is increasingly becoming a collaborator and re-creator in human memory processes in the current era of rapidly advancing AI generative technologies. It challenges the traditional structure of memory that relies on physiological perception and linear narrative, ushering us into an age of algorithm-assisted and data-driven memory reconstruction.

Kidd and McAvoy (2023), in their study of MyHeritage’s “Deep Nostalgia” technology, introduce the concepts of remastered memory and algorithmic nostalgia to describe how algorithms animate photographs to evoke and reshape memory. In the reappearance of visual memory, AI possesses powerful image recognition and narrative generation capabilities. These tools no longer function solely as aids for recollection; instead, they actively engage in memory orchestration and imaginative construction.

A more tangible example can be found in Reminix, which demonstrates how AI can transform static memories into immersive narrative scenes (Reminix, no date). After users upload old photographs, the system uses image recognition technology to reconstruct spatial structures and, through AI modelling, generates navigable 3D spaces. When users “step into” an old photo using VR devices, they are not merely experiencing an enlarged

image. Still, rather than an immersive narrative reconstruction and sensory simulation, memory becomes a virtual experience that can be entered, edited, and reimagined.

However, as previously questioned, when memory spaces are simulated as interactive, editable data environments, is the embodied and emotionally saturated experience diluted?



Figure 4&5: Reminix uses old photos to generate 3D space by algorithms

Source: Reminix - Projects - ECAL.

Although Reminix's 3D spatial reconstructions may faithfully replicate visual and structural elements, they struggle to convey the subtle components that form the depth of real memory, such as the musty smell of damp walls or the creaking of wooden floors underfoot. These unquantifiable sensory details are the irreplaceable essence of memory. While AI-generated memory environments excel in precision, orderliness, and visual clarity, such highly optimised modes of reproduction may also diminish memory's inherent indeterminacy, vagueness, and affective nuance.

In this sense, AI-reconstructed memory spaces are not faithful reproductions of past realities but algorithmically driven re-creations and reconfigurations. These environments often appear more coherent, more ordered—perhaps even more “perfect”—but in doing so, they may lose the serendipity and emotional density intrinsic to original memory. As digital technology increasingly dominates the spatial expression of memory, what we “remember” is no longer the place once inhabited by the body, but a simulated image reconstructed on a screen.

This leads to a series of questions: when AI becomes a co-designer of memory spaces, are we losing a form of primordial embodied experience? Or is it now transforming into an

interface-driven, disembodied landscape of data? Can AI truly reproduce the deepest layers of memory—those defined by intimacy, emotion, and bodily experience?

Memory in AI Simulation: From Emotional Illusion to the Loss of Embodied Presence

Even though AI is increasingly participating in reconstructing human memory, ranging from image recognition and temporal reasoning to narrative generation and 3D modelling, it appears to be becoming a “collaborator of memory.” While technology may simulate a sense of intimacy, whether such simulation can genuinely connect to emotionally resonant memories and reach the deepest layer of remembrance, intimacy remains questionable (Turkle, 2011).

Intimate memories are not easily externalised or translated into “data sets”; they are characterised by subjectivity, affectivity, embodiment, and unpredictability. These memories are often deeply tied to specific spatial and temporal contexts, such as the scent that arises when stepping into a familiar old house, music that evokes adolescent anxiety, or the fleeting gaze captured in a blurry photograph. These moments may be unstable, incomplete, or unclear, yet they are often more truthful and profound than structured digital recollections.

Recent projects like MyHeritage’s Deep Nostalgia (Kidd and Nieto McAvoy, 2023) attempt to resurrect lost loved ones using image generation and motion capture technologies. Similarly, MIT Media Lab’s Re·Live VR project seeks to reconstruct users’ childhood bedrooms through immersive virtual environments, allowing individuals to revisit fragments of their past (Danry *et al.*, 2025). While experiences like Re·Live significantly enhance people’s sense of the past and can evoke intense emotional responses, seemingly “reconstructing” the memory scene on the surface, it still struggles to replicate the spatial structures and embodied perceptions that form the deep foundations of mnemonic intimacy.



Figure 6: Given videos from the past, Re-Live generates a virtual reality space

Source: Overview < Re-Live: Recreating Past Environments as Immersive Walkable Virtual Reality Spaces — MIT Media Lab

On one hand, current AI systems still face limitations in crafting emotionally resonant spatial atmospheres. For instance, Zhang *et al.* (2024) found that while AI-generated architectural imagery can effectively convey positive emotions such as joy, it struggles to express negative emotions, highlighting a broader issue of emotional flattening in AI-generated spaces.

On the other hand, AI-generated memory spaces often prioritise vision-dominant interfaces, thereby sidelining the multiple sensory engagements that ground embodied memory. In architecture and memory studies, scholar Juhani Pallasmaa emphasises that “human memory is fundamentally embodied—it has muscles and bones, not just neurons” (Pallasmaa, 2012). He draws on philosopher Edward Casey’s insight that “body memory is the natural centre of any sensuous account of remembering” (Casey, 1996). In architectural contexts, memory is evoked through spatial tactility, scale relationships, and bodily movement. When AI-generated models fail to respond to this bodily dimension, they risk falling into abstract and surface-level narratives.

Intimate memories often arise from active interactions with physical space. Space is not merely a passive backdrop but an organic memory component. AI-driven “reconstructions” frequently detach memory from this context, compressing it into screen-bound visuals that lack spatial interactivity and temporal depth. Thus, even if AI can generate images with seemingly high emotional intensity, it still touches only the surface of memory. The deeper

layers—rooted in detail, embedded in space, and remembered by the body—remain, for now, beyond the reach of simulation.

The Phenomenological Gap and the Pharmakon: AI's Limits and the Fragility of Memory

In AI-generated memory scenarios, the phenomenological presence becomes deeply questionable. As Heidegger (1962) and Merleau-Ponty (2012) emphasised, phenomenological experience is constructed through the presence of the body, sensory perception, and interaction with the environment. This immediate lived experience cannot be fully recorded or precisely reconstructed. In this sense, memory is not a static image stored in the brain. Still, a holistic recollection evoked through sensory channels—an immediate response of the living body to the world.

However, the content generated by AI is based largely on pattern recognition and probabilistic models. What it recreates is a collage of possible experiences assembled from pre-existing information. It lacks the unpredictability, fragmentariness, and emotional delay characterising embodied memory. As exemplified by large language models such as GPT, these systems do not “remember” or “understand” but perform probabilistic prediction based on context. As Zellers *et al.* (2019) argue, human learning is not about predicting the next word but making sense of the world. Therefore, AI's "reassembling generation" is a simulation behaviour that lacks experiential and contextual depth.

This illustrates the phenomenological gap: AI-generated images or 3D spaces may appear realistic, but are devoid of invisible yet essential dimensions of human memory—emotional residue, bodily responses, temporal flow, and unpredictability. As such, the memory images co-created by AI may offer a form of visual intimacy but fall short of accessing the most fragile and profound dimensions of memory. This technological gap proves that human experience cannot be entirely codified or reproduced.

This dilemma resonates with Bernard Stiegler's (1998) concept of the *pharmakon*—a term denoting both cure and poison. As an external prosthesis of memory, technology helps us record and reconstruct recollections, but simultaneously alters their very nature. The increasingly screen-based, algorithmic, and controlled memory experiences are gradually displacing those filled with sensory delay and tactile presence.

This screen-based memory also flattens perception into a visual surface. Whether it's images generated from prompts or 3D scenes reconstructed from photos and text, the immersion it provides is visually dominant but not fully embodied. We look at them, are moved by them, perhaps even cry—but is this emotional reaction rooted in true resonance, or merely in the aesthetic simulation? The core of intimate memory—embodiment and chance—is diluted or lost in such tightly controlled systems.

Stiegler (1998) also reminds people that memory is about looking back and our capacity to project into the future. When remembering is outsourced to external media, we risk losing the ability to reflect, construct meaning, and build identity through memory.

From this perspective, AI is not the end of memory, but a critical threshold that prompts reflection. It reminds us that memory is not a fully codable process but a continuous, bodily, and emotional becoming. Amid the transitions brought by digitalisation and intelligent systems, we must ask ourselves: what do we truly wish to preserve—images that can be regenerated, or sensations that cannot be restored?

Conclusion

In the post-human era, memory is no longer a private psychological activity or purely biological process—it has evolved into a dynamic system co-produced by humans and technology. The intervention of AI, VR, databases, and algorithms has externalised,

encoded, and reconstructed memory, rendering it more visible and fluid. Memory is no longer confined to internal cognition and emotional experience, but emerges as a hybrid field shaped across physical and virtual spaces.

This essay explores how space serves as an essential container for memory, from embodied experience to digital simulation. Genuine bodily perception and environmental interaction are crucial for maintaining memory's intimacy and emotional richness. With the rise of AI and digital environments, memory locations go beyond physical cultural sites to include immersive virtual spaces. While digital reconstruction expands how memory can be expressed and disseminated, such simulated spaces often lack the richness of embodied experience and the depth of sensory engagement, diluting the "presence" of memory. AI merely reassembles past images and information through data and algorithms, yet lacks the randomness and emotional residue embedded in human experience.

Therefore, as technology continues to reshape the spatiality of memory, this philosophical reflection urges us to reconsider: how can we establish a new balance between intelligent generation and spatial perception? We need not reject AI's role in memory construction, but must remain conscious and self-aware in its use, making room for ambiguity, contingency, imperfection, and the ineffable. Truly profound memories may not emerge at the moment of being "generated," but in the quiet interplay of space, body, and time.

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